Q-Reliability Lab

Computational-basis success can remain perfect while phase-flip noise destroys the target Bell-state relation

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Independent research project

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Classical density-matrix simulation with deterministic finite-shot sampling; no quantum-hardware data.

Outline

1. Motivation and precise research question
2. Phase-flip model and analytic predictions
3. Basis-sensitive success metrics
4. Exact density-matrix sweep
5. Finite-shot experiment and uncertainty
6. Channel–metric sensitivity comparison
7. Reproducibility, limits, and implications
8. Conclusions and future work

Seminar claim

For a Bell-state benchmark, a success definition based only on accepted computational-basis populations cannot certify the relative phase of the state.

Question and model

A Bell-state benchmark under a first-qubit phase-flip channel

Motivation: accepted outcomes can hide state damage

- Bell-state preparation targets $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$.
- A computational-basis task can accept both 00 and 11, reporting $S_Z = P(00) + P(11)$.
- $|\Phi^+\rangle$ and $|\Phi^-\rangle = (|00\rangle - |11\rangle)/\sqrt{2}$ have the same accepted populations.
- Their relative phase differs, so agreement with the intended target differs.

Concrete failure mode

At $p = 1$, the output becomes $|\Phi^-\rangle$:

$$S_Z = 1 \qquad F_{\Phi^+} = 0.$$

Reliability question

Does a perfect task success score actually establish preparation of the desired Bell state?

Bell state and phase-flip channel

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}},$$

$$\rho_+ = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}.$$

$$E_0 = \sqrt{1-p}\, I_4, \qquad E_1 = \sqrt{p}\,(Z \otimes I).$$

- Ordered basis: $(|00\rangle, |01\rangle, |10\rangle, |11\rangle)$.
- Noise acts only after the ideal Bell circuit.
- $p \in [0, 1]$ denotes the channel probability.

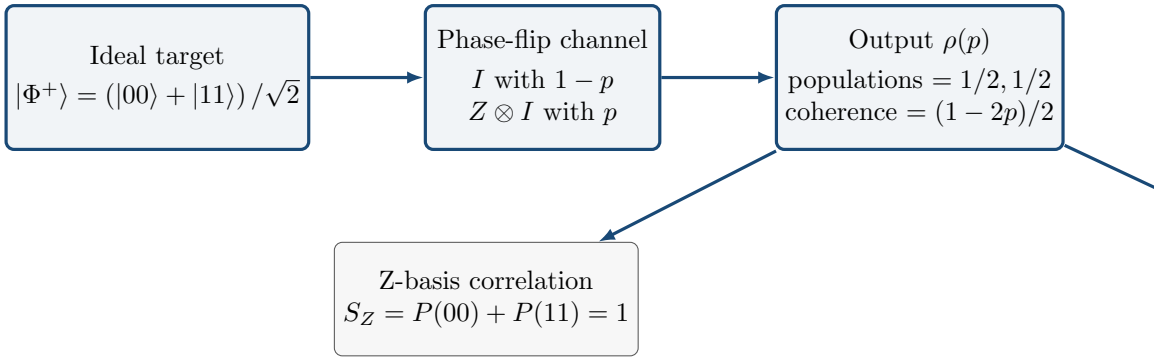
Kraus completeness

$$\sum_k E_k^\dagger E_k = (1-p) I_4 + p(Z \otimes I)^2 = I_4.$$

Physical interpretation

$(Z \otimes I)\,|\Phi^+\rangle = |\Phi^-\rangle$, so the channel mixes two orthogonal Bell states.

What a phase flip changes — and what it leaves untouched



Reading the diagram

The phase flip changes the off-diagonal relation, not the accepted Z-basis populations. The two success metrics therefore answer different physical questions.

Central question and hypothesis

Question

For the stated phase-flip model, which measured quantities remain informative as p changes from zero to one?

Hypothesis

The computational-basis correlated-success metric is structurally insensitive to the phase flip, while fidelity, signed coherence, and an X-basis correlation expose the change.

- Exact sweep: 101 values, $p = 0.00, 0.01, \ldots, 1.00$.
- Finite-shot sweep: five p values, three shot budgets, two bases, 100 repetitions.

Claim boundary

The study reproduces a known measurement-basis limitation as a transparent benchmark. It does not introduce a new noise model or estimate a hardware error rate.

Falsifiable contrast

If S_Z falls with p under this model, the analytic derivation or implementation is incorrect.

Exact analytic baseline

$$\rho(p) = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 1-2p \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1-2p & 0 & 0 & 1 \end{pmatrix}.$$

$$S_Z(p) = 1, \qquad F_{\Phi^+}(p) = 1 - p,$$

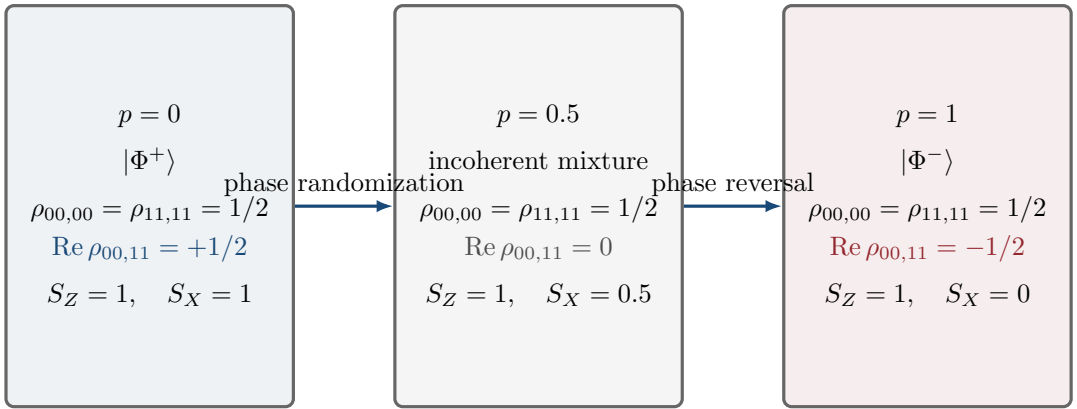
$$\mathrm{Re}\,\rho_{00,11}(p) = \frac{1-2p}{2}, \qquad |\rho_{00,11}(p)| = \frac{|1-2p|}{2}.$$

p	S_Z	F	$\mathrm{Re}\,\rho_{00,11}$
0.00	1.00	1.00	0.50
0.50	1.00	0.50	0.00
1.00	1.00	0.00	-0.50

Endpoint interpretation

$p = 0.5$ removes coherence. $p = 1$ reverses its sign and produces a state orthogonal to the target.

Three exact snapshots: population is unchanged, phase is not



Why this matters

All three states produce the same accepted computational-basis populations. Only a phase-sensitive observable distinguishes the intended Bell relation from a fully reversed one.

Metrics and exact sweep

Success definitions determine which errors a benchmark can observe

Z-basis and X-basis correlated success

$$
\begin{aligned}
S_Z(\rho) &= \rho_{00,00} + \rho_{11,11}, \\
\Pi_X &= |{+}{+}\rangle\langle{+}{+}| + |{-}{-}\rangle\langle{-}{-}|\,, \\
S_X(\rho) &= \mathrm{Tr}(\Pi_X \rho) \\
&= \tfrac{1}{2}(\rho_{00,00} + \rho_{11,11} + \rho_{00,11} + \rho_{11,00}).
\end{aligned}
$$

- Code evaluates S_X from the projector, rather than hardcoding $1 - p$.
- S_Z uses only populations; S_X also uses off-diagonal coherence.

Metric	Information tested
S_Z	Computational-basis correlated populations
S_X	Relative phase through X-basis correlation
Fidelity	Agreement with the specified $\|\Phi^+\rangle$ target
Signed coherence	Direction and size of $\rho_{00,11}$

Structural blindness of computational-basis success

$$\rho_{00,00}(p) = \tfrac{1}{2},$$
$$\rho_{11,11}(p) = \tfrac{1}{2},$$
$$S_Z(p) = \rho_{00,00}(p) + \rho_{11,11}(p) = 1.$$

- The diagonal populations remain fixed under a Z operation on the first qubit.
- Only the cross terms acquire the factor $1 - 2p$.
- Therefore additional Z-basis samples cannot reveal p in this model.

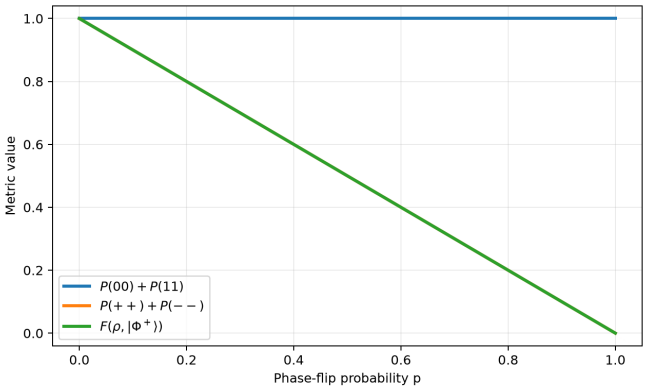
Structural, not numerical

The constancy follows from the channel and measurement projector. It does not result from a coarse probability grid, floating-point rounding, or insufficient shot count.

Benchmark implication

A task score may measure a valid but incomplete property of the output state.

Exact density-matrix sweep

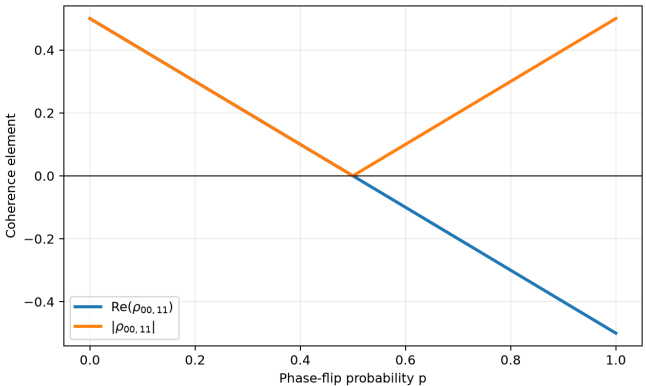


- The exact simulator applies the Kraus map to ρ_+ at 101 values of p.
- S_Z remains equal to one across the entire sweep.
- S_X and fidelity both decrease linearly as $1 - p$.
- The visual agreement reflects shared analytic predictions, checked by parameterized tests.

Conclusion

The selected Z-basis task accepts an output whose target-state fidelity falls continuously from one to zero.

Signed coherence distinguishes loss from phase reversal



- $|\rho_{00,11}|$ decreases to zero at $p = 0.5$ and returns to 0.5 at $p = 1$.
- The magnitude alone cannot distinguish $|\Phi^+\rangle$ from $|\Phi^-\rangle$.
- $\mathrm{Re}\,\rho_{00,11}$ changes sign and records the phase reversal.

Interpretation

At $p = 1$, coherence survives but has the wrong sign for the target state.

Finite-shot evidence

Sampling uncertainty changes precision, not metric sensitivity

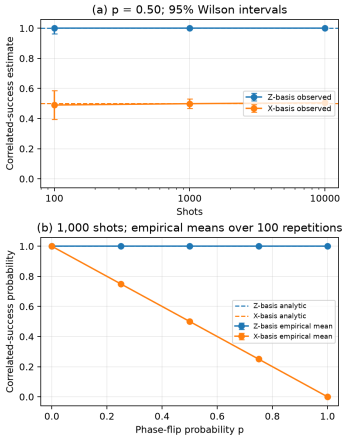
Finite-shot experiment design

Design field	Committed specification
p values	$0, 0.25, 0.50, 0.75, 1.00$
Shot budgets	$100, 1{,}000, 10{,}000$
Measurement bases	Z and X
Repetitions	100 per condition
Root seed	20261030
CSV rows	3,000
Interval	two-sided 95% Wilson

$$K_Z = N,$$
$$K_X \sim \text{Binomial}(N, 1-p),$$
$$\widehat{S}_B = K_B/N.$$

- Each row records its child seed, success count, estimate, analytic value, and interval endpoints.
- These observations are simulated binomial samples, not hardware measurements.

Finite-shot results at $p = 0.50$



- Every Z-basis observation reports correlated success one, for every shot budget.
- X-basis estimates remain close to the analytic value $S_X = 0.5$.
- The X-basis interval narrows as N increases.
- More samples improve an informative estimator; they do not repair an uninformative measurement basis.

Conclusion

Finite sampling introduces uncertainty around S_X but leaves the Z-basis blind spot unchanged.

Wilson intervals quantify sampling precision

N	$\widehat{S}_X$	95% Wilson lower	upper	
100	0.4900	0.3942	0.5865	All rows
1,000	0.4990	0.4681	0.5299	
10,000	0.5053	0.4955	0.5151	

use $p = 0.50$, basis X, repetition 0. The analytic success probability equals 0.5000.

- Wilson intervals remain bounded at all-success and zero-success endpoints.
- Their width reflects finite sample size, not uncertainty about the model parameters.
- A Wilson interval is a frequentist confidence interval, not a posterior probability.

Comparative benchmark

A metric can be blind to one Pauli error while detecting another

Noise-channel and metric sensitivity matrix



- Cells report the maximum absolute metric change over $p \in [0, 1]$.
- S_Z is structurally blind to phase flips.
- S_X is structurally blind to bit flips.
- Every listed metric changes under the specified depolarizing channel.

Conclusion

No single listed metric detects every modeled Pauli noise type; benchmark design must match the error mechanism.

Channel comparison at the sweep endpoints

Channel at $p = 1$	S_Z	S_X	F	$\mathrm{Re}\,\rho$	$\lvert\rho\rvert$
Bit flip	0.00	1.00	0.00	0.00	0.00
Phase flip	1.00	0.00	0.00	-0.50	0.50
Depolarizing	0.50	0.50	0.25	0.00	0.00

ρ denotes the target-basis coherence element $\rho_{00,11}$.

- The bit-flip and phase-flip cases exchange which basis metric remains constant.
- Depolarizing noise uses weights $1 - 3p/4, p/4, p/4, p/4$.
- Its target-fidelity endpoint equals $1/4$, consistent with the maximally mixed state.

Reproducibility and interpretation

Evidence, validation boundary, and claims supported by the model

Reproducibility and engineering evidence

Evidence	Project mechanism
Exact analytics	Parameterized regression tests at five fixed p values
Physical channel checks	Trace, Hermiticity, PSD, Kraus completeness, invalid-input tests
Data regeneration	Deterministic CSV sweeps and committed generated table
Finite-shot repeatability	Fixed root seed and stored child seed per row
Automation	GitHub Actions runs tests, regeneration checks, paper build, Ruff, and CodeQL

- Dependencies for CI are pinned in `requirements-ci.txt`.
- The CI check rejects uncommitted changes to deterministic data artifacts.
- The visual lab implements the same phase-flip metrics and is tested against the Python model.

Reproducibility boundary

These checks validate the stated simulation and artifacts; they do not validate quantum hardware behavior.

Threats to validity and scope limitations

Model limitations

- Classical density-matrix and binomial simulation only.
- One target state, one post-circuit noise location, and three Pauli channels.
- Ideal preparation and measurement in the exact model.

Omitted hardware effects

SPAM error, readout error, crosstalk, calibration drift, coherent control error, and device-specific CNOT noise.

What the study establishes

Within the declared channel and measurement model, the selected Z-basis success metric cannot identify phase-flip strength.

What it does not establish

The study does not measure an IBM or other device error rate, demonstrate a fault-tolerant protocol, or provide a complete process characterization.

Discussion: implications for benchmark design

- A metric can remain numerically stable because its measurement projector excludes the affected degree of freedom.
- More shots reduce uncertainty only for variation that the measurement can observe.
- Fidelity detects mismatch with a chosen target but does not fully characterize a general quantum process.
- Signed coherence adds directional phase information that coherence magnitude discards.

Practical reading of a benchmark score

Before interpreting a high score as reliable state preparation, specify: the target state, accepted outcomes, measurement basis, noise model, and uncertainty model.

Design implication

Use complementary observables when the task specification depends on both population and phase information.

Conclusion

1. The first-qubit phase-flip channel preserves $P(00) + P(11)$ for the Bell-state benchmark, so $S_Z(p) = 1$ across the full sweep.
2. The target fidelity and X-basis correlated success both equal $1 - p$; they expose degradation missed by the computational-basis score.
3. At $p = 1$, the output is the coherent but target-orthogonal state $|\Phi^-\rangle$, yielding $S_Z = 1$ and $F_{\Phi^+} = 0$.
4. Deterministic data, finite-shot intervals, tests, CI, and an interactive visual lab make the benchmark inspectable and reproducible.

Honest contribution

A reproducible quantitative demonstration of a known metric blind spot, framed as a compact reliability benchmark rather than a novel physical discovery.

Future work

- Extend the state family beyond $|\Phi^+\rangle$ and vary the noise location within an explicit circuit.
- Compare coherent over-rotation, amplitude damping, readout error, and correlated multi-qubit noise.
- Add tomography-inspired or randomized-benchmarking-style observables where their assumptions are explicit.
- Run a separately calibrated hardware experiment with documented device, backend date, transpilation, and shot protocol.
- Report hardware uncertainty separately from simulation uncertainty.
- Evaluate whether metric blind spots persist under realistic task-level acceptance rules.

Next empirical step

Hardware data would require a new experiment design; the present simulation supplies analytic controls and regression baselines for that extension.

References



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J. Preskill. Quantum Computing in the NISQ Era and Beyond. *Quantum*, 2:79, 2018. doi:10.22331/q-2018-08-06-79.



C. J. Wood, J. D. Biamonte, and D. G. Cory. Tensor Networks and Graphical Calculus for Open Quantum Systems. *Quantum Information and Computation*, 15(9–10):759–811, 2015.



Z.-J. Liu. *Q-Reliability Lab*. Version 0.1.0, 2026. Repository: `github.com/Raina-ya/q-reliability-lab`.

Backup: output-state derivation

$$\begin{aligned}\mathcal{E}_p(\rho_+) &= (1-p)\rho_+ + p(Z\otimes I)\rho_+(Z\otimes I)\\ &= (1-p)\,|\Phi^+\rangle\langle\Phi^+| + p\,|\Phi^-\rangle\langle\Phi^-|\,.\end{aligned}$$

$$\rho(p) = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 1-2p \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1-2p & 0 & 0 & 1 \end{pmatrix}.$$

- Diagonal entries remain $1/2$.
- The coherence crosses zero at $p = 1/2$.
- The endpoint at $p = 1$ has negative coherence, not zero coherence.

Backup: representative exact sweep values

p	S_Z	S_X	F	$\mathrm{Re}\,\rho_{00,11}$
0.00	1.00	1.00	1.00	0.50
0.25	1.00	0.75	0.75	0.25
0.50	1.00	0.50	0.50	0.00
0.75	1.00	0.25	0.25	-0.25
1.00	1.00	0.00	0.00	-0.50

Cross-check

The table equals the committed CSV values and the analytic relations used in the regression tests.

Backup: deterministic regeneration commands

Exact sweep, figures, and paper table

```
python -m src.phase_error_blind_spot --output experiments/phase_error_sweep.csv
--figures paper/figures --table paper/generated/phase_error_results_table.tex
```

Finite-shot data and figure

```
python -m src.finite_shot_phase_error --output experiments/finite_shot_sweep.csv
--root-seed 20261030
```

```
python -m src.finite_shot_figures --input experiments/finite_shot_sweep.csv --output
paper/figures/finite_shot_detectability.png
```

Channel matrix

```
python -m src.channel_metric_matrix --output experiments/channel_metric_matrix.csv
--figure paper/figures/channel_metric_sensitivity_matrix.png
```